Lab Practical - 3A - FA 2024

Station 1 – Solutions and Dilutions

Commented [1]: Setup near Analytical Balance
Need: Saline solution (to measure density of), transfer pipet,
-- Different Solutions for different sections/Instructors?
Rack of 10 mL volumetric flasks, stoppers, and paper towels

Version A

- Using a dry 10.00 mL volumetric flask, make the necessary measurements and calculate the **density** of your assigned solution
    - Do not transfer liquid into the flask while the flask is on the balance
    - If you get the outside of the flask wet, dry it with a paper towel before putting it on the balance.
- $Density = \frac{mass}{volume}$

Version B

- Use the data table below to calculate the mass-volume % of Solution A, an aqueous solution of NaCl made in a 50.00 mL volumetric flask.

$$Mass\,Volume\,Percent\,\left(\frac{m}{V}\%\right) = \frac{mass\ solute\ g}{mL\ solution} \times 100$$

	Solution A
Mass of 50-mL volumetric flask + stopper (empty)	46.9067 g
Mass of 50-mL volumetric flask + stopper + NaCl	51.9344 g
Mass of 50-mL volumetric flask + stopper + “Solution A”	100.1147 g

Version C

- To make 50.00 mL of “Solution A”, a student weighed 5.0234 g of NaCl. Calculate the molarity of “Solution A”.

The molar mass of NaCl is 58.44277 g/mol

$Molarity\ (M)\ =\ (mol\ solute)/(L\ solution)$

Station 1 - General Lab Knowledge

Version A

- Record the mass of a dry 10 mL volumetric flask
- Remove volumetric flask from balance and fill to the line with DI water
- Record the mass of the full 10 mL flask.
- Calculate the mass of water inside the flask.

Commented [2]: Set up:
-10-mL volumetric flask
- balance
- wash bottle
- transfer pipets
- periodic table

Version B

- Calculate the number of moles of $Ca_3(PO_4)_2$ in 25.3 g.

Version C

- Calculate the molarity of a solution made by dissolving 15.0234 g of NaCl in enough water to make 100.0 mL of solution.

The molar mass of NaCl is 58.44277 g/mol

$$Molarity\ (M) = (mol\ solute)/(L\ solution)$$

Station 2 – General Lab Knowledge

Remember: Record any necessary measurements. Show your work! Check your units, and significant figures!

Version A

- A student just finished making magnesium oxide by reacting Mg with excess oxygen gas from the air.

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

- The MgO that the student made is contained in the vial labeled "A".
  - The mass measurement is recorded on the vial.
- On your report sheet,
  - Assuming the student used 4.4879 g of Mg, determine the **mass due to oxygen** in the product collected in vial "A".

Version B

- A student just finished making magnesium oxide by reacting magnesium with oxygen gas from the air. The student collected their product and put it in a vial labeled "B".

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

- On your report sheet,
  - Determine the mass of MgO in the vial labeled "B". (Show your work!)
    - Assume the mass of the vial is the same as the mass of the empty vial provided.
  - Is this measurement the Actual Yield, Theoretical Yield or Percent Yield?

Version C

- Record the mass of Aluminum foil provided, labeled "C".
- From your measured value, what is the maximum amount of aluminum chloride (in grams) that could be made?

$$2Al(s) + 3Cl_2 \rightarrow \ 2AlCl_3(s)$$

Substance	Molar Mass
Aluminum foil (Al)	26.98 g/mol
Aluminum Chloride ($AlCl_3$)	133.34 g/mol

Commented [CU3]: The setup and supplies for this station is the same as the supplies for the Station 2 for the Alum lab.
Version A and B will use the same vials, C will have the same piece of Al foil.
Reactions changed.

Commented [CU4]: The mass on the vial is 7.4415 g. The student can determine mass of O via subtraction (MgO-Mg), or by stoichiometry (mass O_2). Same answer achieved.

Station 2 - Synthesis of Alum

Remember: Record any necessary measurements. Show your work! Check your units, and significant figures!

Version A

- A student calculated the most Alum they could have made is 8.879 g.
- The Alum that the student made is contained in the vial labeled "A".
  - The mass measurement is recorded on the vial.
- On your report sheet:
  - Calculate the Percent Yield of the reaction.
  - Is this a reasonable value?

$$Percent\ Yield\ (\%)\ =\ (Actual\ Yield)/(Theoretical\ Yield)\ \times 100\%$$

Version B

- A student made the Alum sealed in the vial labeled "B".
- On your report sheet,
  - Determine the mass of the Alum in the vial labeled "B". (Show your work!)
    - Assume the mass of the vial is the same as the mass of the empty vial provided.
  - Is this measurement the Actual Yield, Theoretical Yield or Percent Yield?

Version C

- Record the mass of Aluminum foil provided, labeled "C".
- From your measured value, what is the maximum amount of Alum (in grams) that could be made?

$$2Al + 2KOH + 4H_2SO_4 + 22H_2O \rightarrow 2KAl(SO_4)_2{\cdot}12H_2O + 3H_2$$

Substance	Molar Mass
Aluminum foil (Al)	26.98 g/mol
Alum ($KAl(SO_4)_2{\cdot}12H_2O$)	474.46 g/mol

Commented [5]: Mass measurement should be on the vial

Commented [6]: Setup near Analytical Balance
* Version A needs vial of powder, with mass written on it. (Labeled A)
* Version B needs vial of powder sealed in vial and empty vial. (Both labeled B)
* Version C needs a square of Aluminum Foil,

Station 3 – Determination of Gas Constant R

A student reacted a small piece of magnesium ribbon with hydrochloric acid according to the equation below. The experiment has just completed.

$$\mathbf{Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)}$$

Version A

- Refer to the balanced chemical equation above.
- Calculate the mass of hydrogen gas (in grams) that could be made if a student used a 0.0475 g piece of magnesium.

Version B

- Record the volume of gas collected in this experiment.

Version C

- Record the temperature of the water.
- Record the atmospheric pressure (P_{total}) in the classroom.
- What is the pressure of hydrogen gas in the collection tube (in mm Hg)?
    - $P_{total} = P_{H_2} + P_{H_2O}$

Commented [7]: Setup:
-A Eudiometer set up in a water-filled beaker (inverted) with a measurable amount of gas inside. (Clamped like normal to a bench stand, so it is steady and able to be transported.)
-Thermometer
-CRC handbook
-The weather station should be placed nearby with the inHg to mmHg conversion posted on the side.
- Yellow Sheet
+ Ideal gas law?

Station 3 – Alternate

Commented [8]: Setup:
-A buret with water.
-An Erlenmeyer flask with a thermometer.

Version A

- In Experiment 11a (Synthesis of Alum), a student weighed 0.4276 g of Aluminum foil to begin the experiment.
- Starting with that mass of Aluminum, calculate the maximum amount of Alum (in grams) that could be made? (Use the equation and molar masses listed below.)

$$2Al + 2KOH + 4H_2SO_4 + 22H_2O \rightarrow 2KAl(SO_4)_2{\cdot}12H_2O + 3H_2$$

Substance	Molar Mass
Aluminum foil, Al	26.98 g/mol
Alum, $KAl(SO_4)_2{\cdot}12H_2O$	474.46 /mol

Version B

- A titration has just been completed using the buret at your station.
- On your response sheet, record the final buret reading.

Version C

- On your response sheet, record the temperature of the water using the thermometer provided.
- Then, convert that temperature to Kelvin.

Station 4 - Calorimetry

Version A

1) Add approximately 25 mL of water to a styrofoam cup.
2) Measure the current temperature of water in the styrofoam cup and record it on your response sheet.
3) Put the water into the waste beaker and throw the cup in the trash can.

Version B

1) Add approximately 25-mL of water to a styrofoam cup.
2) Add 1 spoonful of the provided solid to the water.
3) With the thermometer, GENTLY stir the mixture to dissolve the solid while watching the temperature change.
   - Did the temperature go up or down? Record your answer on your response sheet.
   - Was the reaction exothermic or endothermic? Record your answer on your response sheet.
4) Pour your reaction mixture in the provided waste container, wash off the thermometer, and throw your cup in the trash can.

Version C

- Given the data below, calculate the heat absorbed or released by the solution.

$$q = m \times C \times \Delta T$$

Mass of NaOH added	30.25 g
Mass Total Solution (m)	35.0392 g
Specific Heat of Solution (c)	4.184 J/g°C
Change in Temperature (ΔT)	21.1 °C
Heat (q)	

Commented [9]: Setup:
-Place station by sink
-Provide 15 styrofoam cups (per section)
-Thermometer
- Wash Bottle
- Graduated Cylinder capable of measuring 25 mL.
- Citric acid in an unlabled bottle.
- tablespoon measuring spoon
Yellow + Green sheet.
- Waste Container

Commented [10]: - waste container

Station 4: General Lab Knowledge

Version A

- Add approximately 25 mL of water to the provided 50-mL beaker.
- Measure the temperature of the water in the beaker (in degrees Celsius) and record it on your response sheet.
- Put the water into the waste beaker and throw the cup in the trash can.

Commented [11]: Set up:
- 50-mL beaker
- 10-mL graduated cylinder
- wash bottle
- periodic table

Version B

- Record the mass of the provided 50-mL beaker
- Remove it from the balance and add 10 mL of water to the beaker using the provided graduated cylinder.
- Re-weigh the beaker and calculate the mass of water in the beaker.

Version C

- Calculate the number of moles in 13.7 g of $(NH_4)_2CO_3$

Station 5 - Making Measurements

Commented [12]: Setup:
Ruler, Graduated Cylinder, Beaker of water, wash bottle, paper towels.
Rectangular object labeled "A"
Cylindrical object labeled "B"
Any object labeled "C"

Version A

- Determine the volume of object “A” by measuring the dimensions with the provided ruler. Report your answer in cm^3.

$$V = l \times w \times h$$

Version B

- Record (measure) the longest dimension of object “B” in centimeters (cm).
- Convert your measurement to millimeters (mm).

Version C

- Determine the volume of object “C” by displacement of water.
    - Report your answer in mL.
- When time is called, return water to the beaker provided and dry your object and workstation.

Station 6 - Types of Reactions

WEAR GLOVES

All Versions

1) Mix the two products labeled with your version letter in one of the provided clean test tubes (approximately 1 mL of both is fine).
2) On your response sheet:
    a) Record Observations
    b) State the reaction type (choose from list below)
    c) Write a balanced chemical equation for the reaction, complete with phase symbols.
3) Clean Up by placing your waste in the beaker.
    a) Rinse your test tube with water
    b) Place in rack upside down

"Reaction Types":

- No Reaction
- Acid/Base Neutralization
- Gas Evolution
- Precipitation
- Redox
- Combustion

Commented [13]: Setup:
-Set out 6 reactants from Exp 7, with 2 labeled "A", 2 labeled "B", and 2 labeled "C"
*Bottles should also be labeled with the formula of the chemical in them
-Test tubes
-Test tube rack
-Waste Beaker
-Wash bottle
-paper towels
-gloves
-solubility table/green sheet

Commented [CU14]: Ver A: Na2CO3 & CaCl2
Ver B: Na2CO3 & HCl
Ver C: Na3PO4 & BaCl2

Test each to make sure they give strong precipitation and bubbles. Make sure they are concentrated enough that students will see what they need to see.

Station 7 - Determination of Citric Acid in Soda

Commented [15]: Titration setup:
buret, clamp, bench stand, flask "at end point". Fill buret with water to ~15-19 mark

A student has just finished adding NaOH to a flask containing 6-Down Soda and has so far collected the following data.

$$H_3C_6H_5O_7(aq) + 3NaOH(aq) \rightarrow Na_3C_6H_5O_7\ (aq) + 3H_2O(l)$$

Part II- Determination

NaOH Concentration **0.9573 M** (from standardization procedure)

	Trial 1
Volume of "6 Down" Soda	10.00 mL
Initial Buret Reading (NaOH)	1.27 mL
Final Buret Reading (NaOH)	

Version A

- A titration has just been finished.
- On your report sheet:
    - Record the final buret reading of NaOH.
    - Refer to the data table above, calculate the volume of NaOH used in this trial.

Version B

- Refer to the data table above for the chemical equation, molarity of NaOH, and initial volume of NaOH, and volume of soda used.
- If the final buret reading (NaOH) was 17.12 mL, what is the molarity of citric acid ($H_3C_6H_5O_7$) in the "6 down" soda?

$$Molarity\ (M) = (mol\ solute)/(L\ solution)$$

Version C

- On your response sheet, record the final buret reading.

Station 8 - General Lab Knowledge

All Versions

- What are the names of the pieces of lab equipment labeled with your version letter?
  - Partial Credit - If you do not recall the exact name, describe what it is used for (its purpose).

Commented [CU16]: Suggested Equipment:
Bunsen Burner
Volumetric Pipet
Graduated Cylinder
Test Tube
Watch Glass
Erlenmeyer Flask

Station 9 - General Lab Knowledge

Version A

- What is the final molarity of a solution made by diluting 25.00 mL of a 2.5789 M sodium chloride solution to a final volume of 100.00 mL?

Version B

Commented [17]: Graduated cylinder
Flask of water, thermometer

- Record the volume of water in the graduated cylinder
- Record the temperature of the water in the flask.

Version C

- Measure the length of the line on the paper below (in cm) using the ruler provided:

Commented [CU18]: According to Word, this line is 1.52", or 3.86 cm

Station 10x - Empirical Formula

Version A

A student just finished collecting data from the empirical formula lab and got the following data:

Mass Values from Balance

1. *Mass of Empty Crucible + Lid* ___*21.8563*_____ *g*
2. *Mass of Crucible + Lid + Mg* ___*21.9751*_____ *g*
3. *Mass of Crucible + Lid + MgxOy Compound* ___*22.1715*_____ *g (1st weighing)*
   ___*22.1722*_____ *g (2nd weighing)*

- What was the mass of **oxygen** that reacted?

Version B

A student calculates the following:

1. *Mass of Mg used:* ____*0.1524*______ *g*
2. *Mass of the MgxOy compound:* ____*0.2527*______ *g*
3. *Mass of Oxygen in MgxOy compound:* ___*0.1003*______ *g*

- Using the experimental data, what is the mass percent of **oxygen** in the MgxOy sample?

Version C

A student calculates the following:

1. *Mass of Mg used:* ____*0.1495*______ *g*
2. *Mass of the MgxOy compound:* ____*0.2479*______ *g*
3. *Mass of Oxygen in MgxOy compound:* ___*0.0984*______ *g*

- Using the experimental data, what is the mass percent of **magnesium** in the MgxOy sample?

Commented [19]: Alternate: Calculate the moles of oxygen and moles of Mg used.

Station 10y - pH of Household Items

Commented [20]: Setup:
-Equations page with pH equations.
-Small bottle with cabbage juice
-Test tubes with pH Scale in
-Cabbage Juice indicator (Could we use printed colors on test tubes?)
-A product labeled "version C"
-Test tubes & rack to do the reaction in.
-Wash bottle for cleaning
-Waste beaker
-Paper towels
-gloves?

Version A

- Calculate the molarity of hydronium ([H_3O^+]) for the product whose pH is recorded below.

Product Name	[H_3O^+]	[OH^-]	pH	pOH
Vinegar			2.5	

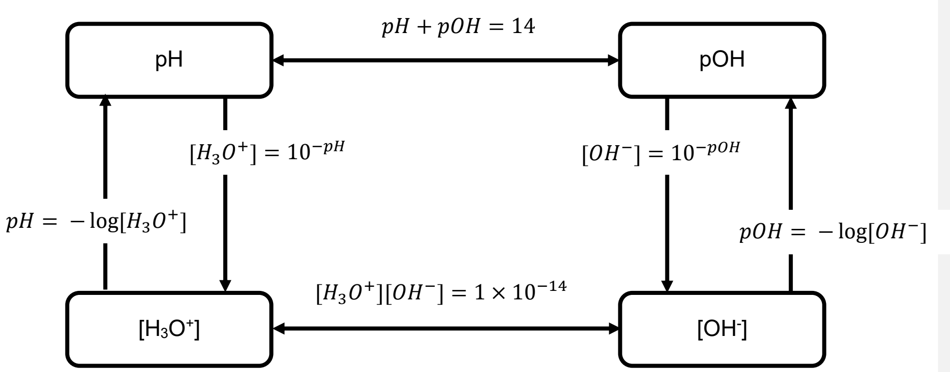
Version B

- Calculate the pOH for the product whose pH is recorded below.

Product Name	[H_3O^+]	[OH^-]	pH	pOH
Vinegar			2.5	

Version C

- Using the materials supplied, determine the pH of the product marked “Version C”.
- When you are done, pour test tube contents into waste. Rinse with water. Put it in the test tube rack upside down.



Station 10z - Electrons in Atoms

Commented [21]: Set up:
-Mercury lamp in power supply (I'm suggesting mercury instead of hydrogen because it is very reliable.)
-Spectroscope in medium clamp on bench stand.(for stability)

Version A

- Using the provided spectroscope, record the color and wavelength (including units) of the highest energy emission from the mercury lamp.

Version B

- Using the provided spectroscope, record the color and wavelength (including units) of the lowest energy emission from the mercury lamp.

Version C

- Using the Rydberg equation (below), calculate the wavelength of the light emitted by the n = 5 to n = 2 electron transition.

$$\frac{1}{\lambda} = \frac{1}{91.13\ nm}\left(\frac{1}{{n_f}^2} - \frac{1}{{n_i}^2}\right)$$

Skill Assessed at Each Station by Version

	Title	A	B	C
Station 1	**NaCl Standard Curve (or Generic Replacement)**	Measurement - Mass Density Calculation Mass calculation (subtraction)	Data Interpretation Calculation-Algebra	Molarity Calculation
Station 2	**Synthesis of Alum**	Data interpretation Calculation - Percent Yield	Measurement-Mass Calculation - Subtraction	Measurement - Mass Stoichiometry (mass to mass)
Station 3	**Stoichiometric Gas Production**	Stoichiometry (mass to mass)	Measurement-Volume	Temperature Measurement Unit Conversion Calculation-Subtraction
Station 4	**Calorimetry**	Measurement-Temperature	Observation	Data interpretation Calculation - Heat
Station 5	**Making Measurements**	Measurement-Length Calculation-volume	Measurement-Length Unit conversion	Measurement-Volume Calculation-Subtraction
Station 6	**Types of Reaction**	Observations, Reaction Classification, Balancing Equation	Observations, Reaction Classification, Balancing Equation	Observations, Reaction Classification, Balancing Equation
Station 7	**Determination of Citric Acid in Soda**	Measurement-Volume Calculation-Subtraction	Data Interpretation Stoichiometry Molarity calculation	Measurement-Volume
Station 8	**General Lab Knowledge**	Name Equipment	Name Equipment	Name Equipment
Station 9	**General Lab Knowledge**	Molarity Calculation	Measurement-Volume Measurement-Temperature	Measurement-Length
Station 10x	**Empirical Formula**	Calculation-Subtraction	Mass %	Mass %
Station 10y	**pH of Household Items**	pH Calculation	pH calculation	Measurement-pH
Station 10z	**Electrons in Atoms**	Measurement-Spectra	Measurement-Spectra	Calculation-Algebra

Station Numbers where Skills are Assessed

	Station Numbers where Skill Assessed		
	A	B	C
Measurement - Mass	1	2	2
Measurement - Volume	7	3, 9	5, 7
Measurement - Temperature	4	9	3
Measurement - Length	5	5	9
Naming Equipment	8	8	8
Stoichiometry Calculation	3	7	2
Molarity Calculation	9	7	1
Calculation involving subtraction of two numbers	1, 7, 10x	1, 2	3, 5
Calculation - Algebra (Multiplication/Division)	1, 2, 5	1	4
Data Interpretation	2	1, 7	4
Unit Conversion		5	3
Observations, Reaction Classification, Balancing Equation	6	6	6